FUNDAMENTALS OF SURVEYING (FS)
CBT EXAM SPECIFICATIONS

Effective Beginning with the January 2014 Examinations

- The FS exam is a computer-based test (CBT). It is closed book with an electronic reference.
- Examinees have 6 hours to complete the FS exam, which contains 110 multiple-choice questions. The 6-hour time also includes a tutorial and a break.
- The FS exam uses the U.S. Customary System (USCS) of units.

Knowledge	Number of Questions
1. **Mathematics**	
A. Algebra, trigonometry, and basic geometry	13–20
B. Spherical trigonometry	
C. Linear algebra and matrix theory	
D. Analytic geometry and calculus	
2. **Basic Sciences**	
A. Geology	5–8
B. Dendrology	
C. Cartography	
D. Environmental sciences	
3. **Spatial Data Acquisition and Reduction**	
A. Vertical measurement	6–9
B. Distance measurement	
C. Angle measurement	
D. Unit conversions	
E. Redundancy	
F. Knowledge and utilization of instruments and methods	
G. Understanding of historical methods and instruments	
4. **Survey Computations and Computer Applications**	
A. Coordinate geometry	19–29
B. Traverse closure and adjustment	
C. Area	
D. Volume	
E. Horizontal curves	
F. Vertical curves	
G. Spirals	
H. Spreadsheets	

5. **Statistics and Adjustments** 6–9
A. Mean, median, mode
B. Variance, standard deviation
C. Error analysis
D. Least squares adjustment
E. Measurement and positional tolerance
F. Relative, network, and positional accuracy

6. **Geodesy** 5–8
A. Basic theory
B. Satellite positioning
C. Gravity
D. Coordinate systems
E. Datums
F. Map projections

7. **Boundary and Cadastral Survey Law** 13–20
A. Controlling elements
B. Gathering and identifying evidence
C. Records research
D. Legal descriptions
E. Case law
F. Riparian rights
G. Public land survey system
H. Metes and bounds
I. Simultaneously created parcels
J. Easements and encumbrances

8. **Photogrammetry and Remote Sensing** 4–6
A. Interpretation and analysis
B. Project and flight planning
C. Quality control
D. Ground control
E. LiDAR

9. **Survey Processes and Methods** 11–17
A. Land development—principles, standards, and regulations
B. Boundary location
C. Mapping, cartography, and topography
D. Construction
E. Riparian surveys
F. Route surveying
G. Control surveys

10. **Geographic Information Systems (GIS)** 5–8
A. Feature collection and integration
B. Database concepts and design
C. Accuracy and use
D. Metadata

11. Graphical Communication and Mapping **6–9**

A. Plans and specifications
B. Contours and slopes
C. Scales
D. Planimetric features and symbols
E. Land forms
F. Digital terrain modeling and digital elevation modeling
G. Survey maps, plats, drawings, and reports

12. Professional Communication **4–6**

A. Oral
B. Written
C. Alternate forms of communication
D. Documentation and recordkeeping

13. Business Concepts **3–5**

A. Contracts
B. Liability and risk management
C. Financial practices
D. Leadership and management principles
E. Personnel management principles
F. Project planning and design
G. Ethics
H. Safety